

1 Introduction

The Large Hadron Collider (LHC) is the world's largest and highest-energy particle accelerator. It was built at the site of European Organization for Nuclear Research (CERN) laboratory in Geneva (Switzerland) during the years 1998 and 2008. The LHC is installed in a 27-kilometer circular tunnel located 50–175 meters

below ground. Inside the accelerator, two particle beams - accelerated nearly to the speed of light - are made to collide. The beams travel in opposite directions in beam pipes, guided around the accelerator ring by a strong magnetic field coming from superconducting electromagnets.

The Compact Muon Solenoid (CMS) is a general-purpose detector at the LHC. It is built around a huge solenoid magnet, which takes the form of a cylindrical coil of superconducting cable that generates a field of around 4 T.

Inside the solenoid, the CMS electromagnetic calorimeter (ECAL) is located. Built of lead tungstate crystals, it is organized into two pseudorapidity (η) regions: the barrel EB ($\eta < 1.479$) and the endcap EE ($1.479 < \eta < 3.0$). The ECAL is surrounded by the hadron calorimeter (HCAL), which covers the pseudorapidity range of up to $\eta = 3.0$. The central tracking detector consists of concentric layers of silicon sensors in a cylindrical volume around 5.4 m long and of a 2.4 m diameter. The large volume of the tracker is required in order to allow significant bending of high energy charged particles (and therefore accurate momentum measurement).

The analysis presented here discusses cross-section measurements for $pp \rightarrow W \rightarrow e\nu$ process with pp collision at 7 TeV center-of-mass.

2 Data

The data used for this analysis were collected by CMS detector and are publicly available from [1]. The data were preselected and contain a subset of the total event information, collected during the 2010 run at the LHC at a centre-of-mass energy of $\sqrt{s} = 7$ TeV. The data correspond to an integrated luminosity of $\mathcal{L} = 36 \text{ pb}^{-1}$. The variables stored in the available format of the dataset are presented in the table below.

Variable	Description
Run	Run number
Event	Event number
pt	Transverse momentum of the electron in units of GeV
eta	Pseudorapidity of the electron
phi	The phi (polar) angle of the electron in radians
Q	Charge of the electron
delEta	The difference in pseudorapidity between the electron track and the associated cluster in the ECAL
delPhi	The difference in phi (in radians) between the electron track and the associated cluster in the ECAL
sigmaEtaEta	The weighted cluster root mean square spread along eta for an electron
HoverE	HCAL energy divided by ECAL energy
isoTrack	The isolation variable of the electron in the tracker
isoEcal	The isolation variable of the electron in the ECAL
isoHcal	The isolation variable of the electron in the HCAL
MET	The missing transverse momentum of the electron in units of GeV

phiMET	The phi angle (in radians) of the missing transverse momentum of the electron
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The total number of events available in the dataset was 100 000. They were collected during several runs (periods of a stable beam, after which the beam is dumped). The number of events preselected from individual runs is shown in Table 1. An event means information registered during a bunch-bunch interaction, recorded for further analysis based on a trigger decision. The number of events preselected from individual runs ranges from a few hundred to a few thousand, with 587 events (run number 165415) being the lowest number of events and 12317 (run number 165567) being the highest.

Run number	Number of events
165567	12317
171446	7273
166049	5508
161312	4666
172868	4645
165467	4548
170876	4152
163817	3792
166565	2546
166408	2420
171484	2416
167676	2404
166514	2398
163376	2350
166782	2253
165514	2251
163659	2241
172949	2240
165633	2219
166034	2067
165121	2044
165537	1776
172822	1600
163402	1456
163758	1420
166512	903
165415	587

Table 1: Number of events stored for this analysis in each run

3 Events selection

The following Figures 1 and 2 show the distribution of electron candidates depending on their pseudorapidity (η), transverse momentum (pt) and charge (Q) values. Distributions are shown separately for the barrel region (EB) and the endcap region (EE). An event was selected if an electron in pseudorapidity range $|\eta| < 1.44$ (EB) or $1.57 < |\eta| < 2.5$ (EE) was found. The transition region $1.44 < |\eta| < 1.57$ was excluded as it leads to lower-quality reconstructed clusters.

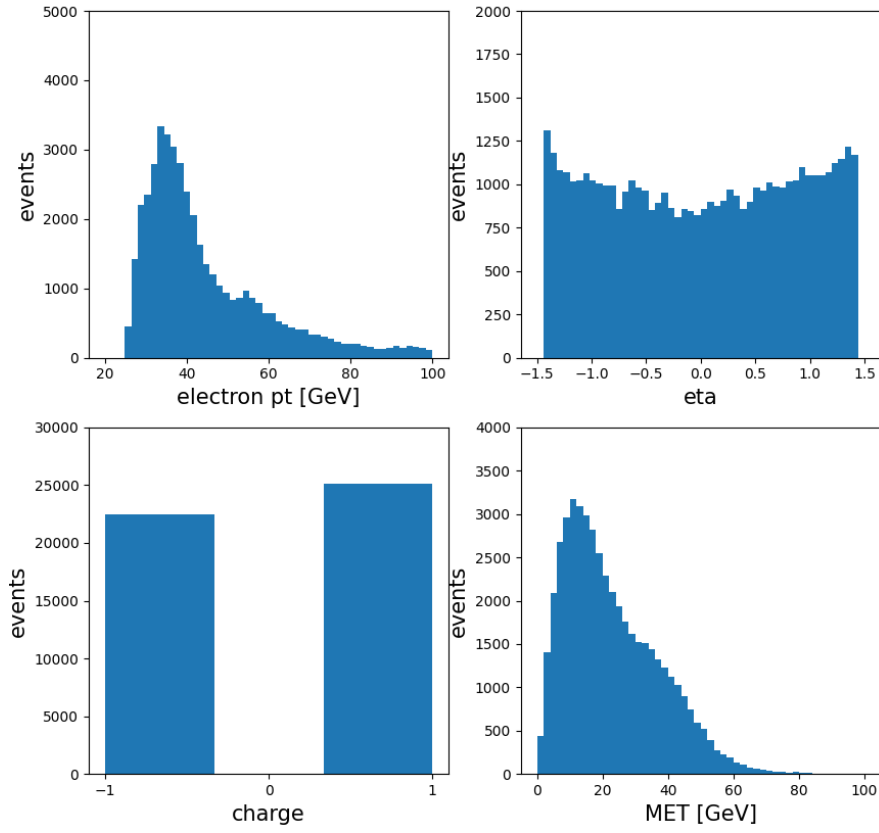


Figure 1: Distributions for electron transverse momenta (p_T), pseudorapidity (η) and charge in EB. There is also shown a distribution for the missing transverse energy of events with electrons in EB.

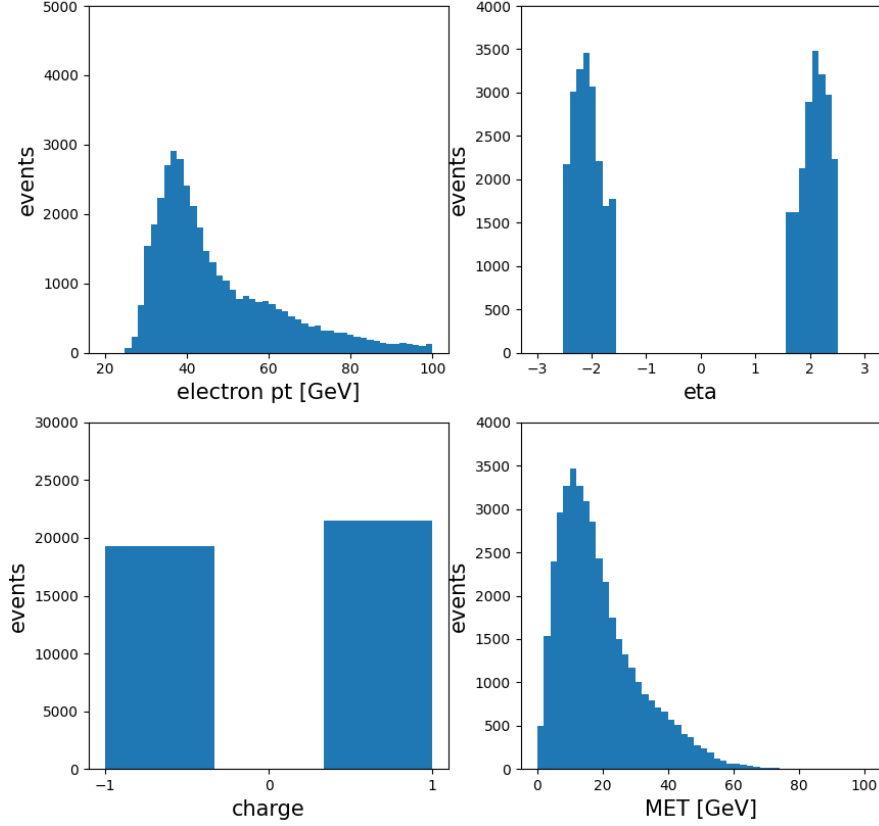


Figure 2: Distributions for electron transverse momenta (p_T), pseudorapidity (η) and charge in EE. There is also shown a distribution for the missing transverse energy of events with electrons in EE.

The average value of the transverse momentum is around 52 GeV in both the barrel region and the endcap region, whereas the average value of the missing transverse energy is close to 23 GeV in EB and 19 GeV in EE. In both regions there is a noticeable charge asymmetry in the number of selected events (for $Q = 1$: 25142 in EB and 21499 in EE, for $Q = -1$: 22429 in EB and 19304 in EE). The intermediate bosons in the $W \rightarrow e\nu$ channel are produced through the quark-antiquark annihilation process in proton-proton collisions. W^+ bosons are produced via $u + \bar{d}$ collision and W^- bosons are produced via $d + \bar{u}$ collisions. The charge asymmetry can be explained by the fact that the colliding protons consist of two u valence quarks and only one d valence quark.

4 Electron reconstruction and identification

Electrons are identified based on matches between clusters of ECAL energy deposits and tracks reconstructed in the silicon tracker. The algorithm used for electron identification can reconstruct clusters containing a large fraction of the energy of the original electron, including energy radiated along its trajectory. The number of particles misidentified as electrons is reduced by putting limitations on the variables $\Delta\eta$, $\Delta\phi$, $\sigma_{\eta\eta}$ (delEta, delPhi, sigmaEtaEta) and the ratio of the hadronic energy H to the electromagnetic energy E (HoverE) measured in a cone of $\Delta R = 0.15$ around the ECAL cluster direction. Electron isolation is obtained by limiting the values of the isolation variables: I_{trk}/p_t , I_{ECAL}/p_t , I_{HCAL}/p_t . The optimization was done for EB and EE separately. The cuts were implemented using requirements listed in Table 2, which follow the published analysis [2]. Figures 3 and 4 show the distributions of the variables listed above, before and after applying the

cuts on the other variables (marked in blue and orange respectively). The shapes of those distributions are similar to the distributions shown in [2]. Analogous distributions cannot be presented for the variables $\Delta\eta$ and $\Delta\phi$, since the available dataset [1] does not contain this information.

Table 3 shows acceptances for cuts on tight identification variables, except the one listed in the columns and for the cuts on all the variables.

	EB	EE
sigmaEtaEta	0.01	0.031
HoverE	0.04	0.025
isoTrack/pt	0.09	0.05
isoEcal/pt	0.07	0.06
isoHcal/pt	0.1	0.025

Table 2: Upper limits for "tight" selection for given variables used in electron identification - EB and EE.

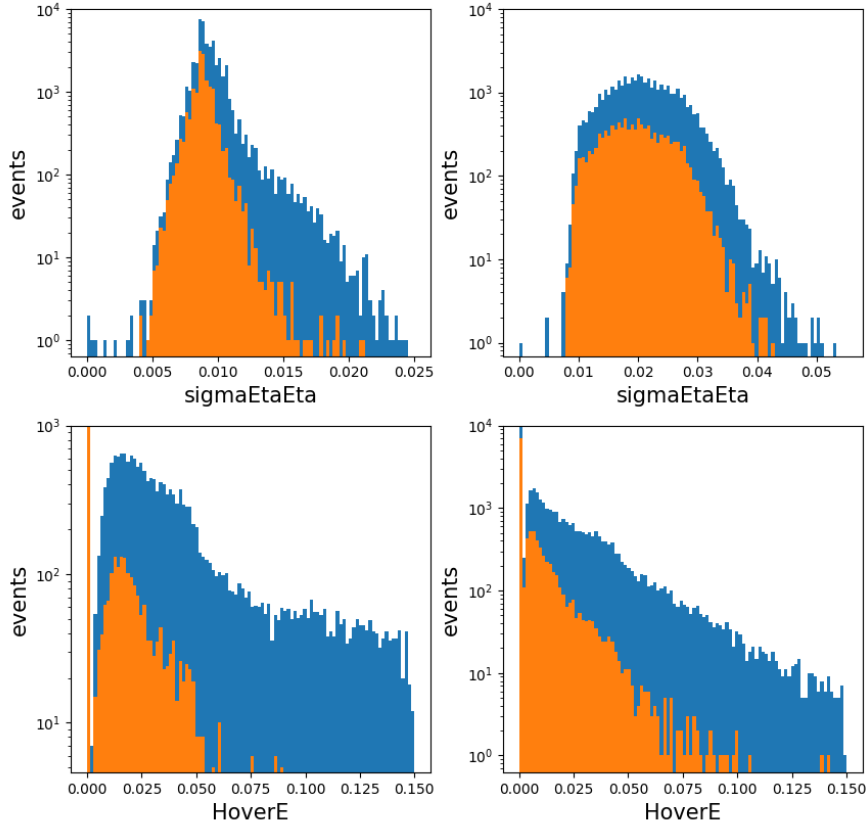


Figure 3: Distributions of the electron identification variables for EB (left) and EE (right)

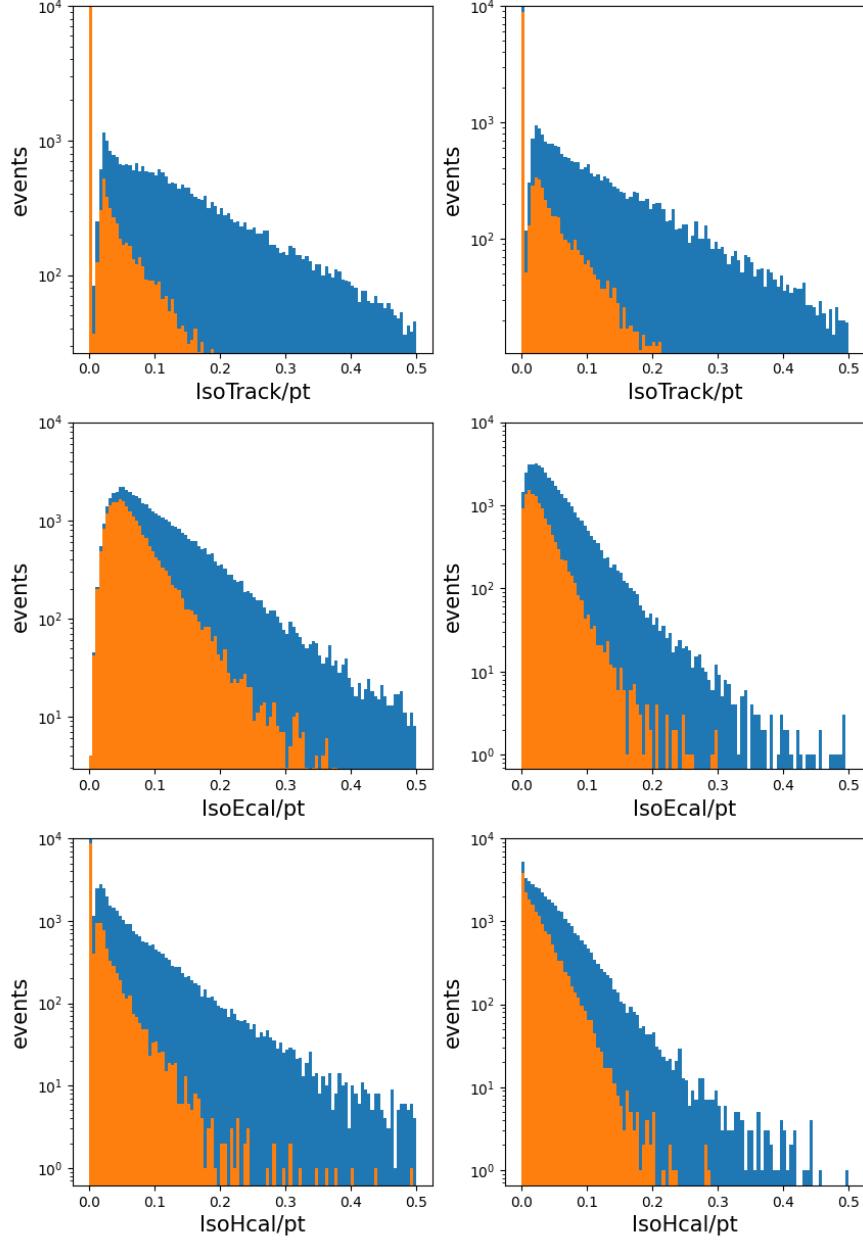


Figure 4: Distributions of the electron isolation variables for EB (left) and EE (right).

Acceptance	sigmaEtaEta	HoverE	isoTrack/pt	isoEcal/pt	isoHcal/pt	Total
EB	0.3237	0.3033	0.3232	0.4698	0.3019	0.2967
EE	0.2708	0.2802	0.3099	0.2981	0.4266	0.2651

Table 3: Acceptance for cuts on identification variables except the one listed in the column and for the cuts on all the variables.

Based on the acceptances presented in Table 3, the most significant cuts for EB and EE were those on the isoEcal/pt variable and the isoHcal/pt variable respectively.

5 Signal and background

After applying all the tight identification criteria, 24 927 events are accepted. For these events, distributions of the kinematic variables: transverse momentum p_t , pseudorapidity η , missing transverse momentum MET and transverse mass m_t are shown in Figures 5 and 6. The transverse mass was calculated using the formula: $m_t = \sqrt{2p_t \cancel{E}_t (1 - \cos(\Delta\phi))}$, where $\cancel{E}_t$ is the MET variable. The distributions after applying tight criteria for MET have two peaks at around 10 and 40 GeV, and the m_t distributions have a peak at around 75 GeV.

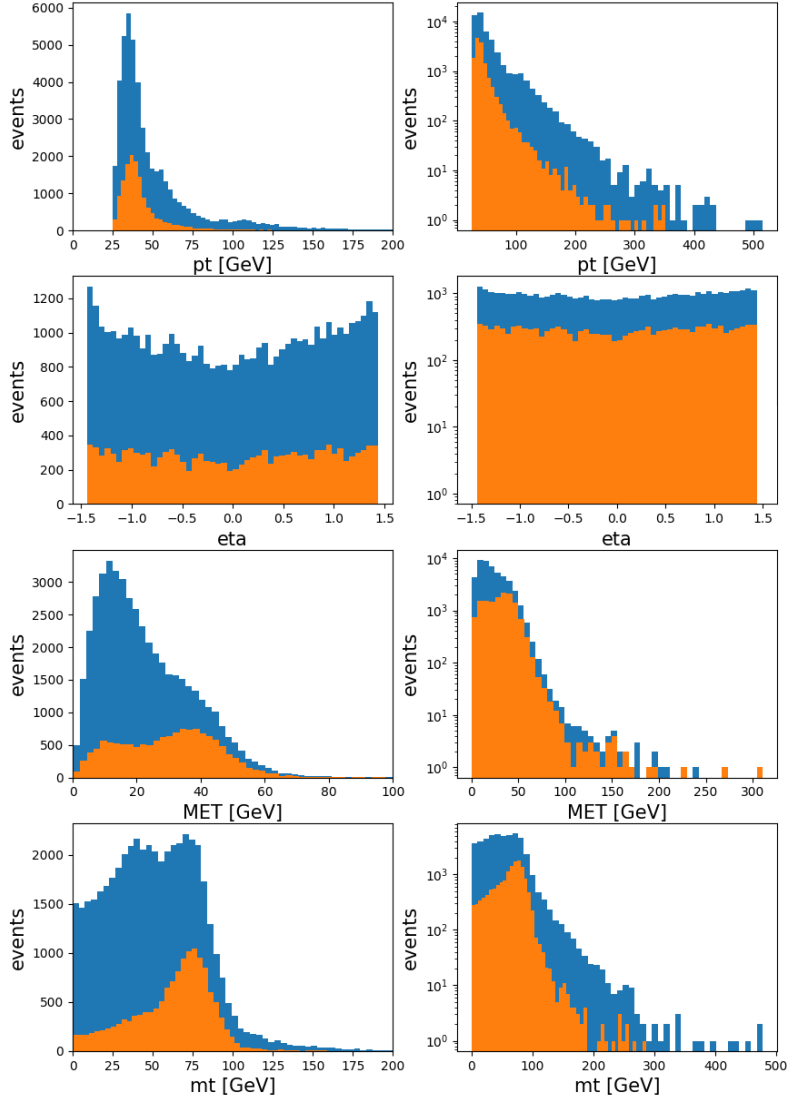


Figure 5: Distributions of the kinematic variables for the EB region - before and after applying tight identification requirements - with a linear scale (left) and on a logarithmic scale (right). Note that for MET and m_t wider range is shown on the logarithmic plot.

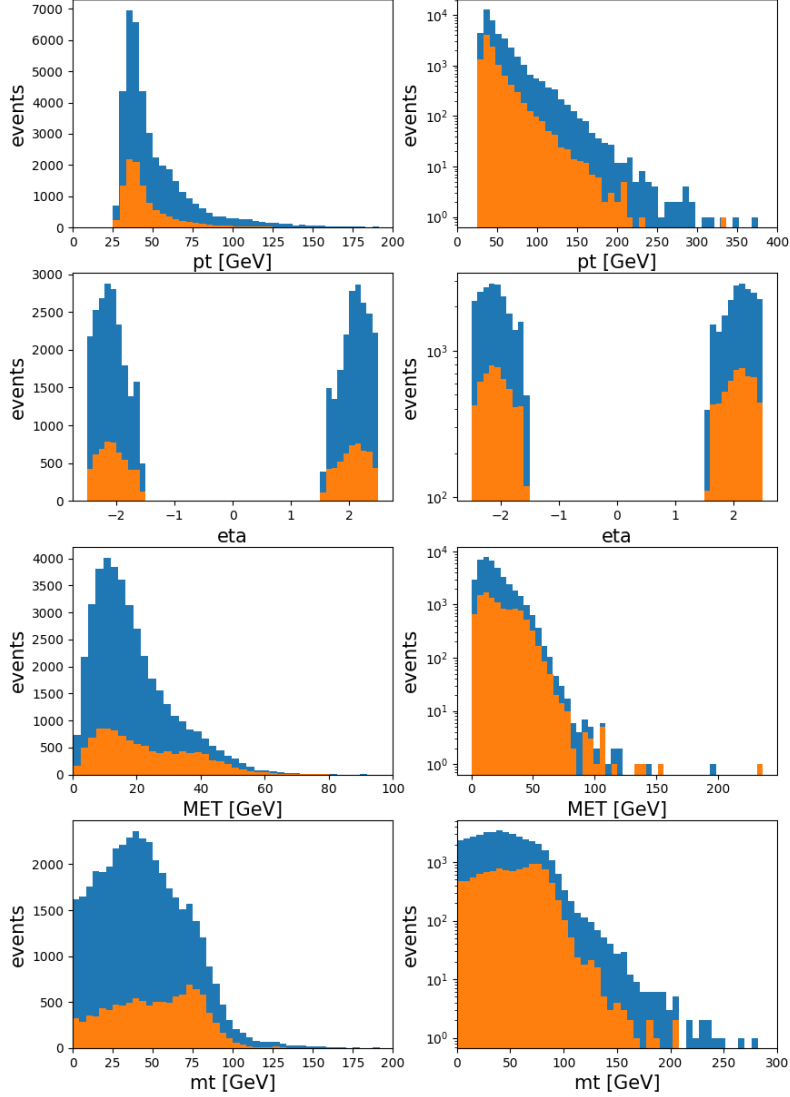


Figure 6: Distributions of the kinematic variables for the EE region - before and after applying tight identification requirements - with a linear scale (left) and a logarithmic scale (right). Note that for MET and mt wider range is shown on the logarithmic plot.

One of the most significant backgrounds in W boson analysis is the QCD background. Although in this analysis other sources of background are neglected, at high MET electroweak backgrounds and the Drell-Yan process also become relevant, leading to contamination levels of the order of 10% [2].

The QCD background is estimated from the data by inverting the identification requirements used to select the signal. The shape of the MET distributions of such events is then fitted in the signal free region of $MET < 25$ GeV for normalisation - according to the publication [2] the $W \rightarrow e\nu$ events are characterized by significant missing transverse energy of at least 25 GeV. The distributions of the kinematic variable MET - for the QCD background in EE and EB - are presented in Figures 7 and 8.

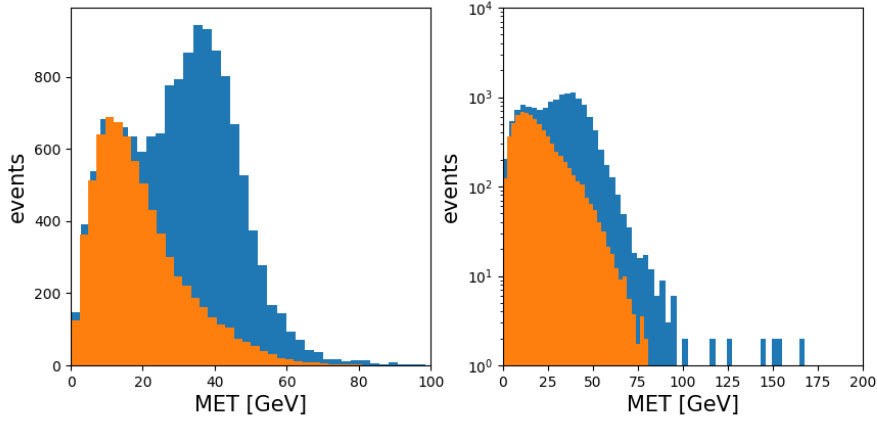


Figure 7: MET distributions in EB - for the selected events (in blue) and for the QCD background (in orange) - on a linear scale (left) and on a logarithmic scale (right). The background distribution was scaled by a factor of 0.22.

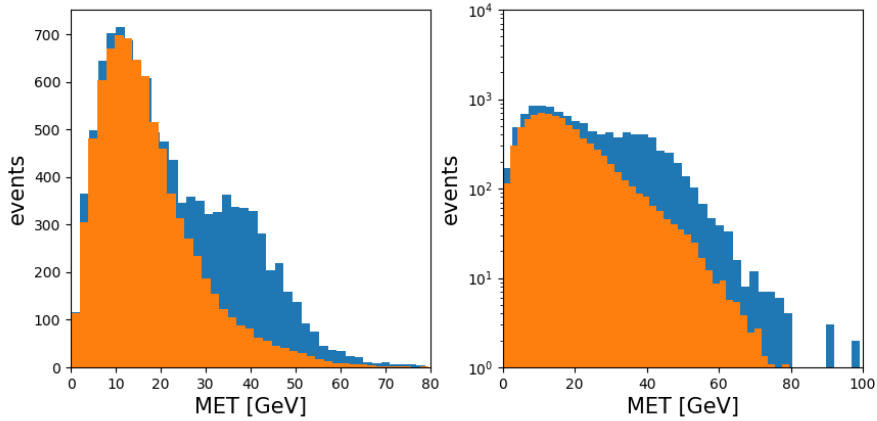


Figure 8: MET distribution in EE - for the selected events (in blue) and for the QCD background (in orange) - on a linear scale (left) and on a logarithmic scale (right). The background distribution was scaled by a factor of 0.27.

The biggest contribution from the QCD background can be observed for the events at MET below ≈ 25 GeV and mt below ≈ 50 GeV (according to the publication [2]). After applying two additional cuts on those variables ($MET > 25$ GeV and $mt > 50$ GeV), the number of selected events is reduced to 8586 in the barrel region and 4030 in the endcap region. After subtracting the QCD background contribution estimated using the normalization from MET distributions in Figures 7 and 8, the number of selected events is 6882 in EB and 2775 in EE - 9657 in total.

6 Cross section

The cross section is defined using the formula:

$$\sigma = \frac{N_{produced}}{\mathcal{L}}$$

where $N_{produced}$ is the number of produced events and $\mathcal{L}$ is the integrated luminosity. In order to estimate $N_{produced}$, the observed number of events N_{signal} obtained by applying the criteria

described in the previous sections needs to be adjusted using two factors - the acceptance A and the efficiency ϵ :

$$N_{produced} = \frac{N_{signal}}{A\epsilon}$$

The acceptance A is the fraction of simulated W events having an ECAL cluster within the ECAL pseudorapidity ranges corresponding to the barrel region and the endcap region with $p_t > 25$ GeV.

The efficiency ϵ is a product of the reconstruction efficiency ϵ_{rec} , the tight selection efficiency ϵ_{tight} and the trigger efficiency ϵ_{trg} . The efficiencies are determined with dedicated measurements using the T&P technique, where one lepton candidate (the “tag”) satisfies the tight selection requirements and the other lepton candidate (the “probe”) is required to pass specific criteria based on a given efficiency.

The values obtained in [2] which are used here: $A = 0.4933 \pm 0.0003$, $\epsilon = 0.798 \pm 0.009$ for the EB region and $\epsilon = 0.67 \pm 0.01$ for the EE region. Applying those factors yields the total number $N_{produced}$: 25878 ± 234 , where the uncertainty was calculated using the propagation of error formula.

The data used for this analysis were preselected - thus, the integrated luminosity of 36 pb^{-1} cannot be used to obtain the final result. We can scale down the luminosity given in the published analysis based on the number of events after removing the QCD background in [2] - which was 136328 ± 386 , and in the given dataset - which is only 9657, as estimated in a previous section. The corresponding luminosity would be equal to around 2.6 pb^{-1} .

The obtained cross section is equal to $9.95 \pm 0.09 \text{ nb}$, comparable to the result published in [2]: 10.48 ± 0.03 (stat.) ± 0.16 (syst.) ± 0.09 (th.) ± 0.42 (lumi.) nb. Within the dataset, an additional uncertainty of $N_{produced}$ stems from the fact that no influence of the electroweak background was taken into account, while in [2] it is estimated to lead to contamination levels of around 11.2%.

Sources

- [1] <http://opendata.cern.ch/record/545>
- [2] <https://arxiv.org/pdf/1107.4789.pdf>
- [3] <https://cds.cern.ch/record/1299116/files/EGM-10-004-pas.pdf>